

FIELD ORIENTED CONTROL OF PERMANENT MAGNET SYNCHRONOUS MOTOR (SENSORLESS)

Shital Sayaji Jadhav^{*1}, Snehal Nandkumar Mahajan^{*2}, Amruta Sudhakar Erandole^{*3}, Jyoti Ashok Ghadage^{*4}

^{*1}Assistant Professor, Department Of Electrical Engineering, Annasaheb Dange College Of Engineering And Technology, Ashta, Maharashtra, India.

^{*2,3,4}Department Of Electrical Engineering, Annasaheb Dange College Of Engineering And Technology, Ashta, Maharashtra, India.

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ABSTRACT

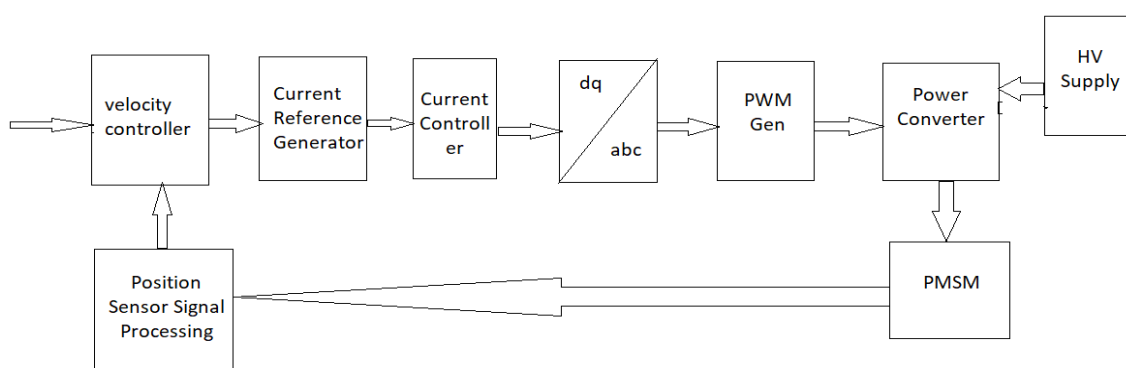
In this paper, the design and simulation of the field oriented controlled (FOC) permanent magnet synchronous motor (PMSM) are studied. PMSMs became important to recent years due to their good dynamic performances and better characteristic features. Mathematical model of PMSM is presented in the paper. FOC scheme, which is one of the most commonly used control methods of industrial applications, is designed by using this model. This control scheme is simulated on Powersim (PSIM) and PMSM based on FOC is analyzed with two different operating conditions. According to simulation results, characteristics of the PMSM are obtained. The error analysis is also presented by comparing these derived characteristics and the experimental results. The relation between the electromagnetic torque and the q-axis current of PMSM is also presented.

Keywords: Control Of PMSM, Field Oriented Control (FOC), Mathematical Model, Permanent Magnet Synchronous Motor (PMSM), Inductance.

I. INTRODUCTION

The "greener" industrial strategies have been heavily taken into account in the past ten years as the market for electronics appliances has increased globally. For purposes of speed, the synchronous permanent magnet motor is a very significant candidate. Industries are the main use-cases for permanent magnet synchronous motors. There are certain benefits like more energy, high - power density and torque, improved performance, and increased dependability. In order to achieve excellent performance of variable speed, a few solutions for field-oriented control have been established. Furthermore, a greater percentage of field-oriented control synchronous permanent magnet motors have sensors than we initially believed. Additionally, installation must be done with extreme care to achieve the desired accuracy. In order to get over these restrictions, a few sensorless rotor position estimate techniques have been developed. The chattering features, however, have a significant impact on the precision and caliber of controllers. Another challenge is added by the frequent usage of the phase-locked-loop (PLL) technology and filter to reduce noise brought on by the speed signals. Recently, a technique for adaptive digital marketing has been utilized to limit chatter, although many viewer parameters' designs have not been fully represented

II. METHODOLOGY



The proposed system aims to achieve the speed of permanent magnet synchronous motor and it completely controlled by vector or flux control method. This framework basically consist of one principle. A sensor is used to recognize the angles of the motor, to analyze how to convert clerk to park transformation and park to clerk transformation.

III. MODELING AND ANALYSIS

In This system we can use AVR Microcontroller Atmega328p-pu. This controller read the analog value of the pot then speed is calibrated pwm signal digital pin 9 then this is connected to ESC electronic speed controller. Using this ESC we can control the speed of the motor. In that ESC 6 hex inverters are there. Then one crystal is there for providing the clock pulse.

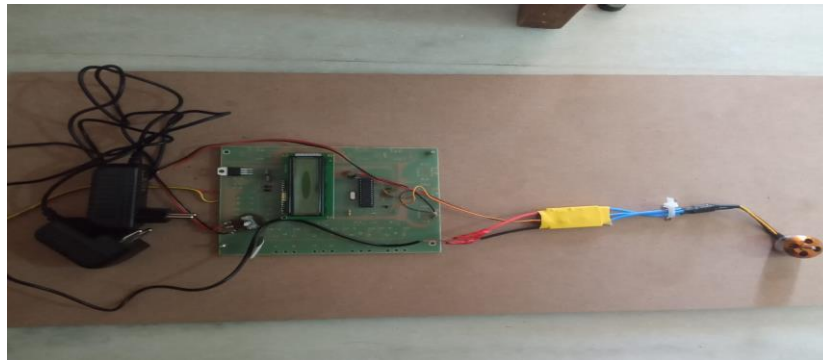


Figure 1: Model Of FOC Of PMSM (Sensorless)

IV. RESULTS AND DISCUSSION

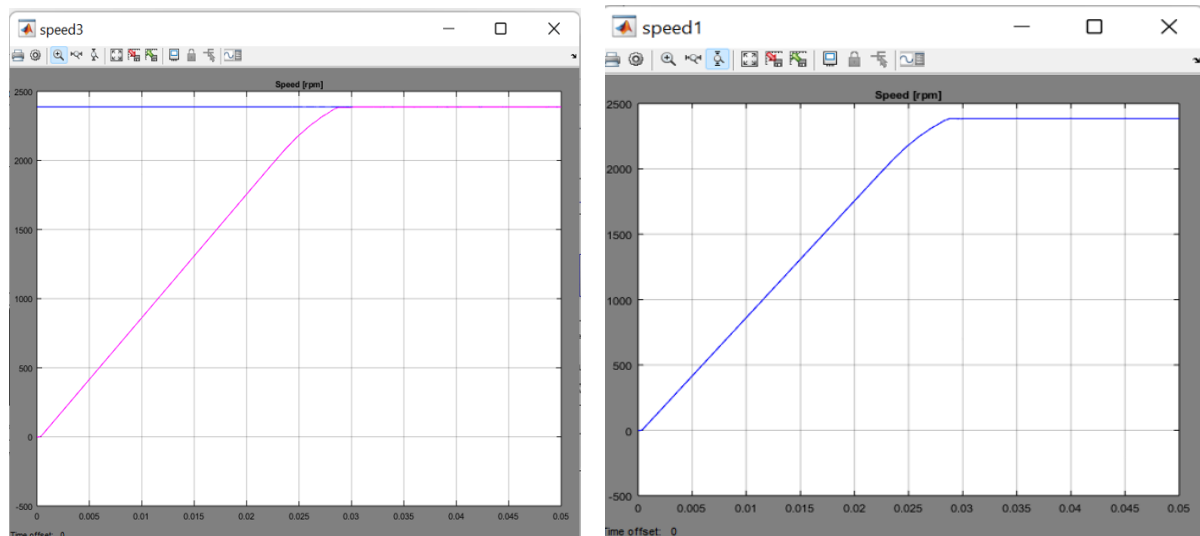


Figure 2: Output

V. CONCLUSION

All In The purpose of this thesis is to assess several sensorless policy measures of a PMSM and determine which of the approaches is most appropriate for a thorough case study. The application of a chain saw made by Husqvarna AB is the subject of the case study. The chainsaw PMSM is currently adept at using a BLDC regulation approach. Husqvarna is looking at the prospect of substituting the current control approach with a sensorless regulating technique tailored for a PMSM in order to enhance the performance and internal control responsibilities of the motor. Together with the specifications of the coordinate transformation functions, there is a mathematical model of the PMSM in the frames of abc, and dq. Also provided is FOC, the original control strategy. To successfully regulate the PMSM, the FOC requires knowledge of the rotor location and speed. According to the notion of sensorless control, an estimator generates the speed and position data.

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VI. REFERENCES

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